



A simple approximation for the bivariate normal integral

Wen-Jen Tsay & Peng-Hsuan Ke

To cite this article: Wen-Jen Tsay & Peng-Hsuan Ke (2021): A simple approximation for the bivariate normal integral, Communications in Statistics - Simulation and Computation, DOI: [10.1080/03610918.2021.1884718](https://doi.org/10.1080/03610918.2021.1884718)

To link to this article: <https://doi.org/10.1080/03610918.2021.1884718>



Published online: 18 Feb 2021.



Submit your article to this journal [↗](#)



Article views: 21



View related articles [↗](#)



View Crossmark data [↗](#)



A simple approximation for the bivariate normal integral

Wen-Jen Tsay and Peng-Hsuan Ke

The Institute of Economics, Academia Sinica, Taipei, Taiwan

ABSTRACT

A simple approximation for the bivariate normal cumulative distribution function based on the error function is derived. We compare the accuracy of our approximation with those of several existing approximation methods and the command in MATLAB. Our approximation performs quite well in the central portion and extreme tails of the distribution. We also apply the proposed method to evaluate the likelihood function of the binary endogenous regressor probit (BERP) model. The simulations indicate that the numerical stability and the accuracy of the proposed approach is promising when conducting the maximum likelihood estimation of the BERP model.

ARTICLE HISTORY

Received 11 July 2020
Accepted 25 January 2021

KEYWORDS

Bivariate normal distribution; Cumulative distribution function; Error function

JEL CLASSIFICATIONS

C16; C25; C63

1. Introduction

This paper provides a fast and accurate formula for approximating the bivariate normal cumulative distribution function (BNCDF), because bivariate normal distribution is one of the most important distributions of applied multivariate analysis and has been widely used in many diverse applications. Artillery fire control is one of the oldest applications of bivariate normal distribution. Pearson (1901, 1903) employs bivariate normal distribution for studying biometric data. Kotz, Balakrishnan, and Johnson (2000, 252–253) provides an overview about the references concerning the historical development of bivariate normal distribution. Genz (2004) reviews new methods for the numerical computation of bivariate normal probabilities.

Bivariate normal distribution also has been extensively used in social science. Greene (2000, 180) claims that achieving a fast and accurate method of computing the BNCDF is a longstanding challenge in applied econometrics. In particular, the BNCDF is required for the probit model with sample selection. Boyes, Hoffman, and Low (1989) apply this sample selection probit model to the problem of bank credit scoring. The BNCDF is necessary for the binary endogenous regressor probit (BERP) model which is popularly used for computing the average treatment effect between the controlled and the treated groups. A notable example is Evans and Schwab (1995) where they apply the BERP model to evaluate the causal effect of attending Catholic school on finishing high school and starting college. Furthermore, the BNCDF is needed when dealing with the binary choice model with binary endogenous regressors in the panel data framework considered in Carrasco (2001).

In this paper we compare our approximation with several existing approximation methods and the command in MATLAB. Our approximation performs very well in the central portion of the BNCDF when compared to the results in Cox and Wermuth (1991) and Lin (1995). In order to check the performance of our approximation for more extreme tails of the BNCDF, we follow the method adopted in Amsler, Schmidt, and Tsay (2019) and Amsler, Papadopoulos, and Schmidt

(2019), where they show that the exact form of the BNCDF is a function of the standard normal cdf under the special case that the second argument of the BNCDF is 0 and the cross-correlation of the two variables is $-1/\sqrt{2}$. As argued in Amsler, Papadopoulos, and Schmidt (2019), this result can be regarded as exact, because it is not an approximation and the normal cumulative distribution function (cdf) can be calculated very accurately over an extremely wide range of argument of the normal cdf. Accordingly, we can use this expression to investigate the relative accuracy of our method with the existing approximations in the extreme tails of the BNCDF.

The remaining parts of this paper are arranged as follows: Sec. 2 presents the approximation formula for the BNCDF. Sec. 3 considers the precision of the proposed method outlined in the following Theorem 1. The finite sample performance of the MLE of the BERP model with our formula is investigated through a Monte Carlo experiment in Sec. 4. We conclude in Sec. 5.

2. Main results

Denote the BNCDF of the bivariate standard normal random vector $(x_1, x_2)^\top$ as:

$$\Phi_2(p, q; \rho) = \int_{x_1 < p} \int_{x_2 < q} \phi_2(x_1, x_2; \rho) \, dx_1 dx_2, \quad (1)$$

where $\phi_2(x_1, x_2; \rho)$ is the bivariate normal probability density function (pdf), and ρ is the correlation coefficient between x_1 and x_2 such that $-1 < \rho < 1$. A closely related quantity is

$$L(p, q; \rho) = \int_p^\infty \int_q^\infty \phi_2(x_1, x_2; \rho) \, dx_1 dx_2. \quad (2)$$

Here, $\Phi_2(\cdot)$ and $L(\cdot)$ are related by the equation:

$$\Phi_2(p, q; \rho) = L(-p, -q; \rho). \quad (3)$$

To compute $\Phi_2(\cdot)$, tables of $L(\cdot)$ have been provided by Lee (1917, 1927), Elderton et al. (1930), and Pearson (1931) among others. Consequently, various approximation methods for calculating $L(\cdot)$ are developed, including Owen (1956), Cox and Wermuth (1991), Lin (1995), and Drezner and Wesolowsky (1990).

Alternative quadrature methods have been proposed to evaluate $\Phi_2(\cdot)$ based on a single integral presentation of $\Phi_2(\cdot)$:

$$\Phi_2(p, q; \rho) = \int_{-\infty}^q \Phi \left[\frac{p - \rho x_2}{(1 - \rho^2)^{1/2}} \right] \phi(x_2) \, dx_2, \quad (4)$$

where $\Phi(\cdot)$ and $\phi(\cdot)$ denote the cdf and the pdf of a standard Gaussian error, respectively. The above single integral representation leads to the possibility of Gauss-Legendre numerical integration. In fact, many popular methods for calculating the BNCDF fall into this category, for example, the 1998 LIMDEP econometric software. Moreover, all the eight algorithms discussed in Terza and Welland (1991) for computing the BNCDF are based on numerical integration procedures. However, we do not know how these procedures perform in the extreme tails of the BNCDF as considered in Amsler, Papadopoulos, and Schmidt (2019).

This paper suggests a simple and accurate approach to approximate the single integral in (4) based on the error function. This implies that the computational burden of our method could be less than that from numerical integration, and our approximation can be easily programmed across different packages. The main idea of this paper is to express the BNCDF in (4) as:

$$\Phi_2(p, q; \rho) = \int_{-\infty}^q \Phi(av + b) \phi(\nu) d\nu, \quad a = \frac{-\rho}{(1 - \rho^2)^{1/2}}, \quad b = \frac{p}{(1 - \rho^2)^{1/2}}. \quad (5)$$

Through (7.1.1) of Abramowitz and Stegun (1970), we denote the error function as:

$$\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt = 2 \int_0^{\sqrt{2}z} \phi(t) dt, \quad (6)$$

where

$$\operatorname{erf}(-x) = -\operatorname{erf}(x). \quad (7)$$

Given that (a, b) in (5) $\in$ finite R when $\rho \in (-1, 1)$, we observe that:

$$\begin{aligned} \Phi_2(p, q; \rho) &= \int_{-\infty}^q \left[\int_{-\infty}^{a\nu+b} \phi(\zeta) d\zeta \right] \phi(\nu) d\nu \\ &= \int_{-\infty}^q \left[\int_{-\infty}^0 \phi(\zeta) d\zeta + \int_0^{a\nu+b} \phi(\zeta) d\zeta \right] \phi(\nu) d\nu \\ &= \int_{-\infty}^q \left[\frac{1}{2} + \frac{1}{2} \operatorname{erf}\left(\frac{a\nu+b}{\sqrt{2}}\right) \right] \phi(\nu) d\nu \\ &= \int_{-\infty}^0 \frac{1}{2} \phi(\nu) d\nu + \int_0^q \frac{1}{2} \phi(\nu) d\nu + \int_{-\infty}^q \frac{1}{2} \operatorname{erf}\left(\frac{a\nu+b}{\sqrt{2}}\right) \phi(\nu) d\nu \\ &= \frac{1}{4} + \frac{1}{4} 2 \int_0^q \phi(\nu) d\nu + \int_{-\infty}^q \frac{1}{2} \operatorname{erf}\left(\frac{a\nu+b}{\sqrt{2}}\right) \phi(\nu) d\nu. \end{aligned}$$

Defining $y = (a\nu + b)/\sqrt{2}$, then we have:

$$\begin{aligned} \Phi_2(p, q; \rho) &= \frac{1}{4} + \frac{1}{4} \operatorname{erf}\left(\frac{q}{\sqrt{2}}\right) + \int_{-\infty}^0 \frac{\sqrt{2}}{2a} \operatorname{erf}(y) \phi\left(\frac{\sqrt{2}y-b}{a}\right) dy \\ &\quad + \int_0^{\frac{aq+b}{\sqrt{2}}} \frac{\sqrt{2}}{2a} \operatorname{erf}(y) \phi\left(\frac{\sqrt{2}y-b}{a}\right) dy. \end{aligned}$$

Again, using the change of variable, $z = -y$, we obtain:

$$\begin{aligned} \Phi_2(p, q; \rho) &= \frac{1}{4} + \frac{1}{4} \operatorname{erf}\left(\frac{q}{\sqrt{2}}\right) - \int_0^{\infty} \frac{\sqrt{2}}{2a} \operatorname{erf}(z) \phi\left(\frac{-\sqrt{2}z-b}{a}\right) dz \\ &\quad + \int_0^{\frac{aq+b}{\sqrt{2}}} \frac{\sqrt{2}}{2a} \operatorname{erf}(y) \phi\left(\frac{\sqrt{2}y-b}{a}\right) dy. \end{aligned}$$

It follows that the analysis centers on approximating the last two terms in the above equation:

$$A = - \int_0^{\infty} \frac{\sqrt{2}}{2a} \operatorname{erf}(z) \phi\left(\frac{-\sqrt{2}z-b}{a}\right) dz, \quad B = \int_0^{\frac{aq+b}{\sqrt{2}}} \frac{\sqrt{2}}{2a} \operatorname{erf}(y) \phi\left(\frac{\sqrt{2}y-b}{a}\right) dy. \quad (8)$$

The cornerstone of this paper is to recognize that the value of $\operatorname{erf}(x)$ in (8) is well approximated by a function, $g(x) = 1 - e^{c_1 x + c_2 x^2}$ for $x \geq 0$, where c_1 and c_2 are chosen to make $g(x)$ as close to $\operatorname{erf}(x)$ as possible. This strategy has never been mentioned in the reviews of Kotz, Balakrishnan, and Johnson (2000) and Genz (2004). It is first employed by Tsay et al. (2013) in deriving the cdf of a composite random variable which is important for the stochastic frontier analysis pioneered in Aigner, Lovell, and Schmidt (1977). The simulations conducted in Tsay et al. (2013) clearly reveals the power of $g(x) = 1 - e^{c_1 x + c_2 x^2}$ in approximating $\operatorname{erf}(x)$. See Tsay et al. (2013) for the details. [Theorem 1](#) presents the theoretical results of this paper.

Theorem 1. $\Phi_2(p, q; \rho)$ can be approximated by $F_{app}(p, q; \rho)$ as such:

1. If $a > 0$ and $aq + b \geq 0$,

$$F_{app}(p, q; \rho) = \frac{1}{2} \left[\operatorname{erf}\left(\frac{q}{\sqrt{2}}\right) + \operatorname{erf}\left(\frac{b}{\sqrt{2}a}\right) \right] + \frac{1}{4\sqrt{1-a^2c_2}} e^{\frac{a^2c_1^2-2\sqrt{2}bc_1+2b^2c_2}{4(1-a^2c_2)}} \times \\ \left[1 - \operatorname{erf}\left(\frac{\sqrt{2}b - a^2c_1}{2a\sqrt{1-a^2c_2}}\right) \right] - \frac{1}{4\sqrt{1-a^2c_2}} e^{\frac{a^2c_1^2+2\sqrt{2}bc_1+2b^2c_2}{4(1-a^2c_2)}} \times \\ \left[\operatorname{erf}\left(\frac{\sqrt{2}q - \sqrt{2}a^2c_2q - \sqrt{2}abc_2 - ac_1}{2\sqrt{1-a^2c_2}}\right) + \operatorname{erf}\left(\frac{a^2c_1 + \sqrt{2}b}{2a\sqrt{1-a^2c_2}}\right) \right].$$

2. If $a = 0$, then $\rho = 0$, by independence of the two variables, we have

$$F_{app}(p, q; \rho) = \Phi(p)\Phi(q).$$

3. If $a > 0$ and $aq + b < 0$,

$$F_{app}(p, q; \rho) = \frac{1}{4\sqrt{1-a^2c_2}} e^{\frac{a^2c_1^2-2\sqrt{2}bc_1+2b^2c_2}{4(1-a^2c_2)}} \left[1 + \operatorname{erf}\left(\frac{\sqrt{2}q - \sqrt{2}a^2c_2q - \sqrt{2}abc_2 + ac_1}{2\sqrt{1-a^2c_2}}\right) \right].$$

4. If $a < 0$ and $aq + b \geq 0$,

$$F_{app}(p, q; \rho) = \frac{1}{2} + \frac{1}{2} \operatorname{erf}\left(\frac{q}{\sqrt{2}}\right) - \frac{1}{4\sqrt{1-a^2c_2}} e^{\frac{a^2c_1^2+2\sqrt{2}bc_1+2b^2c_2}{4(1-a^2c_2)}} \times \\ \left[1 + \operatorname{erf}\left(\frac{\sqrt{2}q - \sqrt{2}a^2c_2q - \sqrt{2}abc_2 - ac_1}{2\sqrt{1-a^2c_2}}\right) \right].$$

5. If $a < 0$ and $aq + b < 0$,

$$F_{app}(p, q; \rho) = \frac{1}{2} - \frac{1}{2} \operatorname{erf}\left(\frac{b}{\sqrt{2}a}\right) - \frac{1}{4\sqrt{1-a^2c_2}} e^{\frac{a^2c_1^2+2\sqrt{2}bc_1+2b^2c_2}{4(1-a^2c_2)}} \times \\ \left[1 - \operatorname{erf}\left(\frac{\sqrt{2}b + a^2c_1}{2a\sqrt{1-a^2c_2}}\right) \right] + \frac{1}{4\sqrt{1-a^2c_2}} e^{\frac{a^2c_1^2-2\sqrt{2}bc_1+2b^2c_2}{4(1-a^2c_2)}} \times \\ \left[\operatorname{erf}\left(\frac{\sqrt{2}q - \sqrt{2}a^2c_2q - \sqrt{2}abc_2 + ac_1}{2\sqrt{1-a^2c_2}}\right) + \operatorname{erf}\left(\frac{-a^2c_1 + \sqrt{2}b}{2a\sqrt{1-a^2c_2}}\right) \right],$$

where a and b are defined in (5), $c_1 = -1.0950081470333$, and $c_2 = -0.75651138383854$.

The derivation of F_{app} is in the [Appendix](#). There are several advantages of using F_{app} in approximating Φ_2 . First, unlike many hybrid numerical integration algorithms, the implementation of F_{app} does not depend on the range of ρ as long as it is not equal to ± 1 , the boundary points. Second, the computation of F_{app} involves the function $\operatorname{erf}(\cdot)$ which can be directly calculated with a standard statistic package. As compared to existing numerical integration algorithms, the BNCDF now can be quickly evaluated with F_{app} .

Third, the technique of approximating $\operatorname{erf}(x)$ with $g(x) = 1 - e^{c_1x+c_2x^2}$ can be generalized to evaluate the trivariate and higher-dimensional normal integrals. Thus, this paper paves a new way to tackle the long-lasting problems of computing multivariate normal integrals.

3. Numerical studies

This section first illustrates the quality of F_{app} in approximating Φ_2 by adapting the design in Table 46.1 of Kotz, Balakrishnan, and Johnson (2000, 276) in [Table 1](#). The command encoded in the GAUSS package, `cdfbvn`, is used as the benchmark for computing Φ_2 . Note that the absolute

Table 1. Accuracy of F_{app} in approximating $\Phi_2(x_1, x_2; \rho)$: probabilities $\times 10^5$.

Method	ρ					
	-0.99	-0.5	-0.1	0.1	0.5	0.99
$x_1 = -2.0 \quad x_2 = -2.0$						
F_{app}	0.00	0.47	29.55	88.23	403.98	1970.88
$\Phi_{2, \text{ GAUSS}}$	0.00	0.32	28.00	87.18	405.29	1971.16
$x_1 = -1.0 \quad x_2 = -1.0$						
F_{app}	0.00	387.54	1951.47	3121.55	6258.61	14498.39
$\Phi_{2, \text{ GAUSS}}$	0.00	378.23	1961.02	3132.02	6251.41	14500.35
$x_1 = 0.0 \quad x_2 = 0.0$						
F_{app}	2257.18	16684.64	23436.40	26563.60	33315.36	47742.82
$\Phi_{2, \text{ GAUSS}}$	2252.67	16666.67	23405.79	26594.21	33333.33	47747.33
$x_1 = 1.0 \quad x_2 = 1.0$						
F_{app}	68267.31	68664.49	70289.96	71460.04	74535.55	82765.70
$\Phi_{2, \text{ GAUSS}}$	68268.95	68647.18	70229.97	71400.97	74520.36	82769.30
$x_1 = 2.0 \quad x_2 = 2.0$						
F_{app}	95449.21	95408.49	95415.05	95473.73	95812.00	97420.09
$\Phi_{2, \text{ GAUSS}}$	95449.97	95450.30	95477.98	95537.15	95855.27	97421.14
$x_1 = -2.0 \quad x_2 = -1.0$						
F_{app}	0.00	17.98	254.25	513.41	1325.70	2275.78
$\Phi_{2, \text{ GAUSS}}$	0.00	14.69	243.34	504.63	1326.62	2275.01
$x_1 = -2.0 \quad x_2 = 0.0$						
F_{app}	0.00	223.92	957.16	1382.32	2093.04	2275.78
$\Phi_{2, \text{ GAUSS}}$	0.00	202.65	923.19	1351.82	2072.37	2275.01
$x_1 = -2.0 \quad x_2 = 1.0$						
F_{app}	0.00	991.27	1826.08	2085.24	2298.98	2275.78
$\Phi_{2, \text{ GAUSS}}$	0.00	948.39	1770.38	2031.67	2260.33	2275.01
$x_1 = -1.0 \quad x_2 = 1.0$						
F_{app}	1368.77	9598.92	12647.44	13844.52	15469.98	15867.16
$\Phi_{2, \text{ GAUSS}}$	1365.17	9614.12	12733.50	13904.50	15487.30	15865.53
$x_1 = 0.0 \quad x_2 = 1.0$						
F_{app}	34134.47	37253.22	41117.56	43016.91	46881.26	50000.00
$\Phi_{2, \text{ GAUSS}}$	34134.47	37260.18	41101.92	43032.56	46874.30	50000.00
$x_1 = 1.0 \quad x_2 = 2.0$						
F_{app}	81857.83	81883.41	82171.27	82432.44	83192.56	84132.84
$\Phi_{2, \text{ GAUSS}}$	81859.46	81874.15	82102.80	82364.09	83186.08	84134.47

Notes: The design of this table follows that of Table 46.1 of Kotz, Balakrishnan, and Johnson (2000, 276). F_{app} is described in Theorem 1. $\Phi_{2, \text{ GAUSS}}$ denotes the value of BNCDF calculated from the code "cdfbvn" of the GAUSS package. The value of cdfbvn is approximately $\pm 5.0 \times 10^{-9}$ for the entire range of arguments.

error of cdfbvn is approximately $\pm 5.0 \times 10^{-9}$ for the entire range of arguments. Table 1 clearly reveals that the performance of F_{app} is excellent in that over half of the differences between F_{app} and cdfbvn are to five decimal places. The worst difference between F_{app} and cdfbvn is only to four decimal places, thus, the worst error of our method shown in Table 1 is to four decimal places. This finding is much better than that found in Table 1 of Cox and Wermuth (1991) and that in Table 1 of Lin (1995) where the worst error within both tables is up to 3 decimal places, indicating the great potential of using our method in approximating bivariate normal integrals.

We also investigate the performance of F_{app} when the correlation coefficient ρ is large by replicating the numerical examples in Table I of Albers and Kallenberg (1994) where ρ is 0.8 and 0.9. Table 2 shows that the worst difference between F_{app} and the value generated from the exact method contained in Table I of Albers and Kallenberg (1994) is not greater than 0.0001 for all configurations considered in that table. The finding strongly supports that F_{app} is better than the approximation methods considered in Table I of Albers and Kallenberg (1994) when ρ is large.

We have shown that our approximation formula is moderately accurate in both absolute and relative terms for non-extreme values of x_1 and x_2 , say $-2 \leq x_1, x_2 \leq 5/2$. In order to check the performance of our approach for more extreme values of x_1 and x_2 , we follow the method adopted in Amsler, Schmidt, and Tsay (2019) and Amsler, Papadopoulos, and Schmidt (2019) to address the issues of evaluating the upper tail and lower tail of the bivariate normal integral.

Table 2. Accuracy of F_{app} in approximating $\Phi_2(-x_1, -x_2; \rho)$: probabilities $\times 10^4$.

(x_1, x_2)									
Method	(0,0)	$(0, -\frac{1}{2})$	(0,-1)	$(\frac{1}{2}, \frac{1}{2})$	$(\frac{1}{2}, 0)$	$(\frac{1}{2}, -\frac{1}{2})$	(1,1)	$(1, \frac{1}{2})$	(1,0)
$\rho = 0.8$									
F_{app}	3975	4693	4944	2185	2778	3021	977	1351	1530
Exact	3976	4692	4944	2186	2778	3022	976	1351	1531
CW(3)	4282	4855	4584	2327	2888	3057	1020	1404	1553
CW(4)	3892	4653	4942	2168	2747	3014	982	1339	1524
AK(6)	3874	4671	4942	2159	2756	3015	1012	1350	1528
AK(7)	3976	4690	4944	2168	2775	3022	959	1346	1530
$\rho = 0.9$									
F_{app}	4281	4883	4992	2453	2969	3077	1155	1496	1580
Exact	4282	4884	4993	2453	2969	3077	1155	1497	1580
CW(3)	4751	4987	5000	2736	3057	3085	1275	1551	1585
CW(4)	4096	4900	4997	2326	2953	3079	1116	1478	1579
AK(6)	4248	4880	4993	2456	2966	3077	1182	1498	1579
AK(7)	4278	4883	4993	2447	2968	3077	1150	1496	1580
(x_1, x_2)									
Method	$(\frac{3}{2}, \frac{3}{2})$	$(\frac{3}{2}, 1)$	$(\frac{3}{2}, \frac{1}{2})$	(2,2)	$(2, \frac{3}{2})$	(2, 1)	$(\frac{5}{2}, \frac{5}{2})$	$(\frac{5}{2}, 2)$	$(\frac{5}{2}, \frac{3}{2})$
$\rho = 0.8$									
F_{app}	349	529	631	98	165	209	22	40	55
Exact	349	530	631	98	165	209	22	41	55
CW(3)	357	548	641	98	170	212	21	41	56
CW(4)	354	527	628	100	165	208	22	41	55
AK(6)	399	539	631	132	175	210	37	46	56
AK(7)	343	527	630	100	165	208	25	41	55
$\rho = 0.9$									
F_{app}	440	614	664	134	203	225	32	53	61
Exact	439	615	663	134	203	225	32	53	61
CW(3)	476	639	667	142	211	226	33	55	62
CW(4)	435	605	663	135	200	225	33	53	61
AK(6)	469	618	663	152	206	225	40	55	61
AK(7)	439	614	663	135	203	225	34	53	61

Notes: The design of this table follows that of Table I of Albers and Kallenberg (1994). F_{app} is described in Theorem 1. The rest results are taken from Table I of Albers and Kallenberg (1994). For example, exact values are computed via Schervish's (1984) procedure, AK(6) denotes the results generated from the first-order approximation method outlined in equation (6) of Albers and Kallenberg (1994), and AK(7) represents the findings from using the second-order approximation approach in equation (7) of Albers and Kallenberg (1994).

Following the notation of Amsler, Schmidt, and Tsay (2019), we denote the Skew Normal density $sn_{\lambda, \sigma}$ as:

$$sn_{\lambda, \sigma}(\varepsilon) = \frac{2}{\sigma} \phi\left(\frac{2}{\sigma}\right) \Phi\left(\frac{\lambda \varepsilon}{\sigma}\right), \quad \sigma > 0. \quad (9)$$

We then define its cdf as

$$P_{\lambda, \sigma}(q) = P(\varepsilon \leq q) = \int_{-\infty}^q sn_{\lambda, \sigma}(\varepsilon) d\varepsilon. \quad (10)$$

Amsler, Schmidt, and Tsay (2019) prove that,

$$P_{1,1}(q) = \Phi(q)^2. \quad (11)$$

Moreover, Amsler, Papadopoulos, and Schmidt (2019) quote the finding in Azzalini and Capitanio (2014), 34, eq. (2.34), that

Table 3. Evaluations based on $2\Phi_2(q, 0; \rho)$.

q	Exact SN-CDF	MATLAB-mvncdf	F_{app}
	$\Phi(q)^2$ (for comparison)		
-8	3.87003e-31	4.78899e-25	6.17948e-27
-6	9.73356e-19	9.69319e-19	1.12000e-16
-4	1.00306e-09	1.00307e-09	5.15653e-09
-2	5.17567e-04	5.17569e-04	6.40737e-04
-1	2.51714e-02	2.51715e-02	2.55884e-02
0	2.50000e-01	2.50000e-01	2.50166e-01
1	7.07861e-01	7.07861e-01	7.08060e-01
2	1 - 4.49826e-02	1 - 4.49827e-02	1 - 4.53635e-02
4	1 - 6.33423e-05	1 - 6.33415e-05	1 - 7.51737e-05
6	1 - 1.97318e-09	1 - 1.97318e-09	1 - 3.92369e-09
8	1 - 1.24419e-15	1 - 1.24419e-15	1 - 6.55036e-15

Notes: The results from $\Phi(q)^2$ and MATLAB-mvncdf are copied from Table 1c of Amsler, Papadopoulos, and Schmidt (2019).

$$P_{\lambda,1}(q) = 2\Phi_2(q, 0; \rho), \quad \rho = \frac{-\lambda}{\sqrt{1 + \lambda^2}}. \quad (12)$$

Combining the results in (13) and (14), Amsler, Papadopoulos, and Schmidt (2019) show that

$$P_{1,1}(q) = \Phi(q)^2 = 2\Phi_2\left(q, 0; \frac{-1}{\sqrt{2}}\right).$$

In other words, the exact form of the BNPDF is a function of the standard normal cdf under the special case that $x_2 = 0$ and $\rho = -1/\sqrt{2}$. Since this normal cdf function can be evaluated very accurately as shown in Amsler, Papadopoulos, and Schmidt (2019), so that we can use this expression to investigate the accuracy of our approximation and the existing methods.

Table 3 displays the values of our approximation and those of the command MATLAB mvncdf, and $\Phi(q)^2$. Our approximation and the exact result of $\Phi(q)^2$ are closed-form expressions, apart from the need to evaluate the normal cdf and error function. Note that the results of MATLAB and $\Phi(q)^2$ are copied from Amsler, Papadopoulos, and Schmidt (2019) for comparison. Indeed, Amsler, Papadopoulos, and Schmidt (2019) consider a variety of different software, commercial or open source for evaluating $\Phi_2(x_1, 0, -1/\sqrt{2})^2$. They document these different routines do not necessarily give the same answers in the tails.

Table 3 shows that the value of our approximation and those of MATLAB command are quite close in absolute but not always in relative terms in the tails of the BNPDF when compared to the exact value of $\Phi(q)^2$. For example, for $q = -1$, the value of $\Phi(q)^2$ is 0.251714, and F_{app} is 0.255884, so that they are close in both absolute and relative terms. However, for $q = -8$, $\Phi(q)^2$ is 3.87003e-31, $F_{app} = 6.17948e-27$, and the value of MATLAB is 4.78899e-25, so that they are close in absolute term but not in relative term. Since our approximation can be programmed across different packages, the promising performance as evident in the above simulations reveals its potential usefulness for potential applications.

In addition to showing the accuracy of F_{app} , we also compare the computational cost of F_{app} with that of MATLAB mvncdf command under the specifications in Table 3. We conduct 10,000 replications for evaluating each bivariate normal integral in Table 3 with both approaches. The means of the times used for each method are recorded for comparison. We find the average computational time for MATLAB mvncdf is 0.000256327 second, but it is 1.34e-5 second for F_{app} when $q = -8$. When $q = 8$, the elapsed time is 0.000255215 second for MATLAB mvncdf command, while it is 1.12e-5 second for F_{app} . On average, F_{app} is about 20 times faster than MATLAB mvncdf command when we take all the specifications in Table 3 into account.

4. Simulations for BERP model

This section investigates the finite sample performance of the MLE of the BERP model using the results in [Theorem 1](#). Evans and Schwab (1995) argue that the selection bias encountered in program evaluation is tremendously important. They claim that, if students with more ability or students from families that place a higher value on education are more likely to attend Catholic schools, then single-equation models that treat school choice as an exogenous variable would overstate the effects of a Catholic school education. This implies that an appropriate model must take this endogeneity into account. Because both the outcome measures and the treatment variable are dichotomous under the set-up of Evans and Schwab (1995), this study falls into the framework of the probit model with binary endogenous regressor:

$$\begin{aligned} y_1 &= 1 [x_1^\top \beta_1 + \alpha_1 y_2 + u_1 > 0], \\ y_2 &= 1 [z_2^\top \delta_2 + v_2 > 0], \end{aligned} \quad (13)$$

where $1(\cdot)$ is the indicator function taking the value one if the statement in the bracket is true and zero otherwise, the correlation coefficient between u_1 and v_2 is ρ_1 with $|\rho_1| < 1$, and (u_1, v_2) is independent of x_1, z_2 and distributed as bivariate standard normal with mean zero.

Wooldridge (2002, 477) explains that the effect of y_2 is the primary interest to the literature. Under this set-up, the important average treatment effect (for a given value of x_1) is calculated as $\Phi(x_1^\top \beta_1 + \alpha_1) - \Phi(x_1^\top \beta_1)$.

It is well-known that the probit estimation of y_1 based on the observations of y_2 and those of x_1 is inconsistent for β_1 and α_1 when $\rho_1 \neq 0$. Joint estimation for the two equations in (13) is inevitable. However, Wooldridge (2002, 478) documents that the maximum likelihood estimation (MLE) of the BERP model is nontrivial. Some two-step procedures thus are proposed as alternatives. Unfortunately, Freedman and Sekhon (2008) show that the usual Heckman two-step procedure should not be used for the BERP model, and likelihood method is preferred. As a consequence, the methodology derived in this paper serves to suggest a simple and accurate approximation method for evaluating the likelihood function of the BERP model. The following simulations clearly show that the bias, mean-squared-error (MSE), and the numerical stability of the MLE based on our proposed method is very promising.

In this paper we show how the parameters in (13) can be easily and accurately estimated with the approximated likelihood function suggested in this paper. For ease of exposition, we represent all the exogenous variables contained in x_1 and z_2 as w . With the independently and identically distributed (iid) observations from $i = 1, \dots, n$, the log-likelihood function for the BERP model consists of four parts:

$$\begin{aligned} \ln L &= \sum_{i=1}^n y_{1i} y_{2i} [\ln f(y_{1i} = 1, y_{2i} = 1 | w_i)] \\ &+ \sum_{i=1}^n (1 - y_{1i}) y_{2i} [\ln f(y_{1i} = 0, y_{2i} = 1 | w_i)] \\ &+ \sum_{i=1}^n y_{1i} (1 - y_{2i}) [\ln f(y_{1i} = 1, y_{2i} = 0 | w_i)] \\ &+ \sum_{i=1}^n (1 - y_{1i}) (1 - y_{2i}) \ln [f(y_{1i} = 0, y_{2i} = 0 | w_i)], \end{aligned} \quad (14)$$

$f(y_1, y_2 | w)$ in (14) is the joint distribution of (y_1, y_2) given w and can be expressed as:

$$f(y_1, y_2 | w) = P(y_1 | y_2, x_1) P(y_2 | z_2). \quad (15)$$

It is easy to figure out that

$$P(y_2 = 1|z_2) = \Phi(z_2^\top \delta_2), \quad P(y_2 = 0|z_2) = 1 - \Phi(z_2^\top \delta_2). \quad (16)$$

By (15.55) and (15.56) of Wooldridge (2002), we have

$$P(y_1 = 1|y_2 = 1, x_1) = \frac{1}{\Phi(z_2^\top \delta_2)} \int_{-z_2^\top \delta_2}^{\infty} \Phi \left[\frac{x_1^\top \beta_1 + \alpha_1 y_2 + \rho_1 v_2}{(1 - \rho_1^2)^{1/2}} \right] \phi(v_2) dv_2, \quad (17)$$

and

$$P(y_1 = 1|y_2 = 0, x_1) = \frac{1}{1 - \Phi(z_2^\top \delta_2)} \int_{-\infty}^{-z_2^\top \delta_2} \Phi \left[\frac{x_1^\top \beta_1 + \alpha_1 y_2 + \rho_1 v_2}{(1 - \rho_1^2)^{1/2}} \right] \phi(v_2) dv_2, \quad (18)$$

respectively.

Because $P(y_1 = 0|y_2 = 1, x_1) = 1 - P(y_1 = 1|y_2 = 1, x_1)$ and $P(y_1 = 0|y_2 = 0, x_1) = 1 - P(y_1 = 1|y_2 = 0, x_1)$, the log-likelihood function can be computed with the above findings as:

$$\begin{aligned} \ln L &= \sum_{i=1}^n y_{1i} y_{2i} \left\{ \ln \int_{-z_{2i}^\top \delta_2}^{\infty} \Phi \left[\frac{x_{1i}^\top \beta_1 + \alpha_1 y_{2i} + \rho_1 v_{2i}}{(1 - \rho_1^2)^{1/2}} \right] \phi(v_{2i}) dv_{2i} \right\} \\ &\quad + \sum_{i=1}^n (1 - y_{1i}) y_{2i} \ln \left\{ \Phi(z_2^\top \delta_2) [1 - P(y_{1i} = 1|y_{2i} = 1, x_{1i})] \right\} \\ &\quad + \sum_{i=1}^n y_{1i} (1 - y_{2i}) \left\{ \ln \int_{-\infty}^{-z_{2i}^\top \delta_2} \Phi \left[\frac{x_{1i}^\top \beta_1 + \alpha_1 y_{2i} + \rho_1 v_{2i}}{(1 - \rho_1^2)^{1/2}} \right] \phi(v_{2i}) dv_{2i} \right\} \\ &\quad + \sum_{i=1}^n (1 - y_{1i}) (1 - y_{2i}) \ln \left\{ [1 - \Phi(z_2^\top \delta_2)] [1 - P(y_{1i} = 1|y_{2i} = 0, x_{1i})] \right\}. \end{aligned} \quad (19)$$

The difficult parts in (19) center on evaluating the following two terms:

$$\int_{-z_2^\top \delta_2}^{\infty} \Phi \left[\frac{x_1^\top \beta_1 + \alpha_1 y_2 + \rho_1 v_2}{(1 - \rho_1^2)^{1/2}} \right] \phi(v_2) dv_2, \quad \int_{-\infty}^{-z_2^\top \delta_2} \Phi \left[\frac{x_1^\top \beta_1 + \alpha_1 y_2 + \rho_1 v_2}{(1 - \rho_1^2)^{1/2}} \right] \phi(v_2) dv_2. \quad (20)$$

The first term in (20) is equivalent to

$$\int_{-\infty}^{\infty} \Phi \left[\frac{x_1^\top \beta_1 + \alpha_1 y_2 + \rho_1 v_2}{(1 - \rho_1^2)^{1/2}} \right] \phi(v_2) dv_2 - \int_{-\infty}^{-z_2^\top \delta_2} \Phi \left[\frac{x_1^\top \beta_1 + \alpha_1 y_2 + \rho_1 v_2}{(1 - \rho_1^2)^{1/2}} \right] \phi(v_2) dv_2. \quad (21)$$

The task is thus simplified to evaluate

$$\int_{-\infty}^{-z_2^\top \delta_2} \Phi \left[\frac{x_1^\top \beta_1 + \alpha_1 y_2 + \rho_1 v_2}{(1 - \rho_1^2)^{1/2}} \right] \phi(v_2) dv_2 = \int_{-\infty}^q \Phi(av + b) \phi(v) dv, \quad (22)$$

where

$$a = \frac{\rho_1}{(1 - \rho_1^2)^{1/2}}, \quad b = \frac{x_1^\top \beta_1 + \alpha_1 y_2}{(1 - \rho_1^2)^{1/2}}, \quad (23)$$

and v is a standard Gaussian error. Thus, the log-likelihood function of the BERP model can be easily evaluated with the results in Theorem 1. To calculate the integral $\int_{-\infty}^{\infty} \Phi(av + b) \phi(v) dv$ in (22), we can plug a large number for q in Theorem 1. Indeed, $q = 20$ is employed throughout the following Monte Carlo experiment. The resulting finite sample performance of the proposed

Table 4. Comparing F_{app} with GAUSS command for BERP models: $\rho_1 = 0.6$.

n	β_{10}	β_{11}	α_1	δ_{20}	δ_{21}	ρ_1
F_{app}						
1000	0.0019	-0.0012	-0.0045	-0.0015	0.0032	-0.0018
2000	0.0076	-0.0024	-0.0092	-0.0024	-0.0028	0.0015
4000	0.0027	-0.0020	-0.0003	-0.0013	-0.0016	-0.0015
8000	0.0021	-0.0007	-0.0006	-0.0021	-0.0012	-0.0012
GAUSS						
1000	0.0085	0.0003	-0.0138	-0.0012	0.0019	0.0024
2000	0.0076	-0.0017	-0.0099	-0.0024	-0.0027	0.0027
4000	0.0003	-0.0024	0.0027	-0.0009	-0.0025	-0.0010
8000	0.0012	0.0006	-0.0011	-0.0021	-0.0020	0.0012
F_{app}						
1000	0.1693	0.0570	0.2027	0.0469	0.0584	0.1024
2000	0.1209	0.0393	0.1422	0.0331	0.0417	0.0724
4000	0.0804	0.0275	0.0968	0.0240	0.0300	0.0504
8000	0.0563	0.0203	0.0694	0.0178	0.0229	0.0354
GAUSS						
1000	0.1700	0.0570	0.2039	0.0468	0.0584	0.1027
2000	0.1211	0.0392	0.1429	0.0332	0.0424	0.0725
4000	0.0799	0.0274	0.0959	0.0240	0.0305	0.0495
8000	0.0569	0.0205	0.0702	0.0179	0.0232	0.0353

Notes: All the results are based on 500 replications. Bias is computed as the true parameter values minus the average estimated values. The simulated data are defined in (24) and (25). F_{app} is defined in Theorem 1. GAUSS denotes the value of BNCDF calculated from cdfbvn of the GAUSS package.

method is excellent in terms of bias and MSE of the approximated maximum likelihood estimator.

The design of Monte Carlo experiment follows closely with that in Freedman and Sekhon (2008). We consider a set of experiments as follows:

$$\begin{aligned} y_{i1}^l &= \mathbf{1}[\beta_{10} + \beta_{11}x_{i11}^l + \alpha_1 y_{i2}^l + u_{i1}^l > 0], \\ y_{i2}^l &= \mathbf{1}[\delta_{20} + \delta_{21}z_{i21}^l + v_{i2}^l > 0], \end{aligned} \quad i = 1, 2, \dots, n, \quad (24)$$

where l denotes the l -th replication of the data, the regressor x_{i11} and z_{i21} are both drawn from $N(0, 1)$, and the correlation coefficient between x_{i11} and z_{i21} is 0.4. The true parameter values considered for the simulations are:

$$\xi = (\beta_{10}, \beta_{11}, \alpha_1, \delta_{20}, \delta_{21}, \rho_1)^\top = (-1, 0.75, 0.5, 0.5, 1, \rho_1)^\top, \quad \rho_1 = \{0.3, 0.6\}. \quad (25)$$

Note that Freedman and Sekhon (2008) only consider the cases with $\rho_1 = 0.6$.

We estimate ξ with the following transformation function:

$$\xi = (\beta_{10}, \beta_{11}, \alpha_1, \delta_{20}, \delta_{21}, \rho_1)^\top = \left(\beta_{10}, \beta_{11}, \alpha_1, \delta_{20}, \delta_{21}, \frac{1 - e^{\tilde{\rho}_1}}{1 + e^{\tilde{\rho}_1}} \right)^\top = \kappa(\tilde{\xi}), \quad (26)$$

where

$$\tilde{\xi} = (\beta_{10}, \beta_{11}, \alpha_1, \delta_{20}, \delta_{21}, \tilde{\rho}_1)^\top \quad (27)$$

are the parameters really estimated when conducting the MLE. In order to create a realistic simulation scheme, the inverse function of the preceding transformation function calculated at the true parameter value plus an extra (6×1) random vector generated from $N(0, 1/3)$ is used as the initial values for the MLE procedure, i.e., the initial value of $\tilde{\xi}$ is:

$$\tilde{\xi}_0 = \tilde{\xi} + N(0, 1/3).$$

All the programs are written in GAUSS at four relatively large sample sizes, $n = 1000, 2000, 4000$, and 8000. Two hundred additional values are generated in order to obtain random starting values.

Table 5. Comparing F_{app} with GAUSS command for BERP models: $\rho_1 = 0.3$.

n	β_{10}	β_{11}	α_1	δ_{20}	δ_{21}	ρ_1
F_{app}						
1000	0.0008	-0.0007	-0.0039	-0.0010	0.0038	0.0026
2000	0.0028	-0.0020	-0.0055	-0.0022	-0.0004	0.0048
4000	0.0012	-0.0010	0.0002	-0.0021	0.0001	0.0021
8000	0.0008	-0.0007	0.0003	-0.0013	-0.0006	0.0009
GAUSS						
1000	0.0039	0.0002	-0.0092	-0.0008	0.0031	0.0045
2000	0.0014	-0.0019	-0.0048	-0.0015	-0.0009	0.0049
4000	0.0016	-0.0006	-0.0007	-0.0015	-0.0008	0.0028
8000	0.0004	0.0002	0.0001	-0.0010	-0.0010	0.0021
F_{app}						
1000	0.1540	0.0582	0.2006	0.0453	0.0589	0.1227
2000	0.1091	0.0401	0.1394	0.0326	0.0438	0.0889
4000	0.0736	0.0294	0.0959	0.0245	0.0316	0.0617
8000	0.0524	0.0209	0.0697	0.0176	0.0233	0.0439
GAUSS						
1000	0.1534	0.0582	0.1996	0.0454	0.0586	0.1228
2000	0.1074	0.0399	0.1386	0.0330	0.0441	0.0879
4000	0.0764	0.0298	0.0993	0.0245	0.0309	0.0635
8000	0.0525	0.0212	0.0705	0.0173	0.0233	0.0435

Notes: All the results are based on 500 replications. Bias is computed as the true parameter values minus the average estimated values. The simulated data are defined in (24) and (25). F_{app} is defined in Theorem 1. Exact denotes the value of BNPDF calculated from cdfbvn of the GAUSS package.

The optimization algorithm used to implement the MLE is the quasi-Newton algorithm of Broyden, Fletcher, Goldfarb, and Shanno (BFGS) contained in the GAUSS MAXLIK library. The maximum number of iterations for each replication is 100. The first 500 replications with normal convergence status are recorded for numerical analysis. Furthermore, the exact method encoded in the GAUSS package, cdfbvn, is again used in this section as the benchmark for implementing the BERP model. The results are contained in Tables 4 and 5 for the specifications $\rho_1 = 0.6$ and $\rho_1 = 0.3$, respectively.

Tables 4 and 5 clearly show that our method is computationally efficient in that it handles the simulations with a large sample size of 8000 and 500 replications easily. The finite sample performance of the MLE based on F_{app} in Theorem 1 is very promising, because the MSE of the maximum likelihood estimator is found to decrease with the increasing sample size n , revealing that the maximum likelihood estimator possesses a well-defined asymptotic behavior. In addition, the corresponding bias is very small when $\rho_1 = 0.6$. The preceding results remain qualitatively intact when $\rho_1 = 0.3$.

More importantly, we find that the MSE generated from F_{app} in Tables 4 and 5 are almost identical to those obtained from the numerical method, cdfbvn, in GAUSS package. This finding provides a direct evidence that the MLE of the BERP model can be easily and accurately carried out with the analytical approximation F_{app} proposed in this paper. Furthermore, during the simulation experiments, the failure of normal convergence from our method occurs much less than that from using the approach in the GAUSS package, showing the stability of using our analytic formula for empirical applications.

5. Conclusions

We propose a simple and moderately accurate approximation method for the bivariate normal cumulative distribution function based on the error function. Our method is better than the approaches considered in Cox and Wermuth (1991) and Lin (1995) in the central portion of the BNPDF. We also find that our approximation is much better than the other approximation formulae considered in Table I of Albers and Kallenberg (1994) where ρ is large in the similar setting.

In this paper we also address the issues of evaluating the upper tail and lower tail of the bivariate normal integral by following the method adopted in Amsler, Schmidt, and Tsay (2019) and Amsler, Papadopoulos, and Schmidt (2019). We find that the values of our approximation are quite close in absolute but not always in relative terms in the tails of the BNCDF when compared to the exact value of $\Phi_2(x_1, 0, -1/\sqrt{2})^2$.

We also apply the proposed method to approximate the likelihood function of the BERP model in order to demonstrate that it can be easily and accurately estimated with the approximation method suggested in this paper, even though Wooldridge (2002, 478) documents that MLE is nontrivial for the BERP model. The simulations clearly reveal that the performance of the maximum likelihood estimator based on our method is satisfactory. All these observations point to the potentials of our methodology for many empirical applications.

Acknowledgements

We would like to thank Professor Peter Schmidt of Michigan State University, I-Lin Ho of Academia Sinica, and the reviewers and Editors of *Communications in Statistics – Simulation and Computation* for their valuable comments and suggestions. We also thank Ping-Shou Zhong of the Iowa State University for the suggestion about the presentation of case 2 of Theorem 1.

Appendix: Proof of Theorem 1

This appendix illustrates the derivation of F_{app} in Theorem 1. For ease of exposition, let us emphasize two equations, (7.1.1) and (7.4.32), of Abramowitz and Stegun (1970) for later applications:

$$\begin{aligned} \operatorname{erf}(z) &= \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt = 2 \int_0^{\sqrt{2}z} \phi(t) dt, \\ \int e^{-(kx^2+2mx+n)} dx &= \frac{1}{2} \sqrt{\frac{\pi}{k}} e^{\frac{m^2-kn}{k}} \operatorname{erf}\left(\sqrt{k}x + \frac{m}{\sqrt{k}}\right) + C, \quad k \neq 0, \end{aligned}$$

where C denotes some finite constant. Moreover, the results in (8) are also replicated here:

$$A = - \int_0^\infty \frac{\sqrt{2}}{2a} \operatorname{erf}(z) \phi\left(\frac{-\sqrt{2}z-b}{a}\right) dz, \quad B = \int_0^{\frac{aq+b}{\sqrt{2}}} \frac{\sqrt{2}}{2a} \operatorname{erf}(y) \phi\left(\frac{\sqrt{2}y-b}{a}\right) dy. \quad (8)$$

For expositional purposes, we divide the analysis into five parts, i.e., $(a > 0, aq + b \geq 0)$, $(a = 0, aq + b \geq 0)$, $(a \geq 0, aq + b < 0)$, $(a < 0, aq + b \geq 0)$, and $(a < 0, aq + b < 0)$.

For the case $(a > 0, aq + b \geq 0)$,

$$\begin{aligned} A &= - \int_0^\infty \frac{\sqrt{2}}{2a} \operatorname{erf}(z) \phi\left(\frac{-\sqrt{2}z-b}{a}\right) dz \\ &\approx - \int_0^\infty \frac{1}{\sqrt{2a}} (1 - e^{c_1 z + c_2 z^2}) \frac{e^{\frac{1}{2}\left(\frac{-\sqrt{2}z-b}{a}\right)^2}}{\sqrt{2\pi}} dz \\ &= - \frac{1}{2a\sqrt{\pi}} \left[\int_0^\infty e^{-\left(\frac{1}{a^2}z^2 + 2\frac{b}{\sqrt{2}a^2}z + \frac{b^2}{2a^2}\right)} dz - \int_0^\infty e^{-\left(\frac{1-a^2c_2}{a^2}z^2 + 2\frac{\sqrt{2}b-a^2c_1}{2a^2}z + \frac{b^2}{2a^2}\right)} dz \right]. \end{aligned}$$

Using the results in (7.4.32) of Abramowitz and Stegun (1970), A can be well approximated by

$$A \approx \frac{-1}{4} + \frac{1}{4} \operatorname{erf}\left(\frac{b}{\sqrt{2a}}\right) + \frac{1}{4\sqrt{1-a^2c_2}} e^{\frac{a^2c_1^2-2\sqrt{2}bc_1+2b^2c_2}{4(1-a^2c_2)}} \left[1 - \operatorname{erf}\left(\frac{\sqrt{2}b-a^2c_1}{2a\sqrt{1-a^2c_2}}\right) \right].$$

We can likewise approximate the value B for the case $(a > 0, aq + b \geq 0)$ as:

$$B \approx \frac{1}{4} \left[\operatorname{erf} \left(\frac{q}{\sqrt{2}} \right) + \operatorname{erf} \left(\frac{b}{\sqrt{2a}} \right) \right] - \frac{1}{4\sqrt{1-a^2c_2}} e^{\frac{a^2c_1^2 + 2\sqrt{2}bc_1 + 2b^2c_2}{4(1-a^2c_2)}} \times \\ \left[\operatorname{erf} \left(\frac{\sqrt{2}aq - \sqrt{2}a^3c_2q - \sqrt{2}a^2bc_2 - a^2c_1}{2a\sqrt{1-a^2c_2}} \right) + \operatorname{erf} \left(\frac{a^2c_1 + \sqrt{2}b}{2a\sqrt{1-a^2c_2}} \right) \right].$$

Combining the preceding results about A and B , we note that $\Phi_2(p, q; \rho)$ can be well approximated by $F_{app}(p, q; \rho)$ for the case $(a > 0, aq + b \geq 0)$ as:

$$F_{app}(p, q; \rho) = \frac{1}{2} \left[\operatorname{erf} \left(\frac{q}{\sqrt{2}} \right) + \operatorname{erf} \left(\frac{b}{\sqrt{2a}} \right) \right] + \frac{1}{4\sqrt{1-a^2c_2}} e^{\frac{a^2c_1^2 - 2\sqrt{2}bc_1 + 2b^2c_2}{4(1-a^2c_2)}} \times \\ \left[1 - \operatorname{erf} \left(\frac{\sqrt{2}b - a^2c_1}{2a\sqrt{1-a^2c_2}} \right) \right] - \frac{1}{4\sqrt{1-a^2c_2}} e^{\frac{a^2c_1^2 + 2\sqrt{2}bc_1 + 2b^2c_2}{4(1-a^2c_2)}} \times \\ \left[\operatorname{erf} \left(\frac{\sqrt{2}q - \sqrt{2}a^2c_2q - \sqrt{2}abc_2 - ac_1}{2\sqrt{1-a^2c_2}} \right) + \operatorname{erf} \left(\frac{a^2c_1 + \sqrt{2}b}{2a\sqrt{1-a^2c_2}} \right) \right].$$

Following the preceding procedures, we derive the other four cases without difficulty. The details of the proof can be provided upon request. Combining the results from the aforementioned five cases, we prove [Theorem 1](#).

References

- Abramowitz, M., and I. A. Stegun. 1970. *Handbook of mathematical functions*. New York, NY: Dover.
- Aigner, D., C. A. K. Lovell, and P. Schmidt. 1977. Formulation and estimation of stochastic frontier production function models. *Journal of Econometrics* 6 (1):21–37. doi:[10.1016/0304-4076\(77\)90052-5](https://doi.org/10.1016/0304-4076(77)90052-5).
- Albers, W., and C. M. Kallenberg. 1994. A simple approximation to the bivariate normal distribution with large correlation coefficient. *Journal of Multivariate Analysis* 49 (1):87–96. doi:[10.1006/jmva.1994.1015](https://doi.org/10.1006/jmva.1994.1015).
- Amsler, C., A. Papadopoulos, and P. Schmidt. 2019. Evaluating the cdf of the Skew Normal distribution. *Empirical Economics*, forthcoming.
- Amsler, C., P. Schmidt, and W. J. Tsay. 2019. Evaluating the cdf of the distribution of the stochastic frontier composed error. *Journal of Productivity Analysis* 52 (1–3):29–35. doi:[10.1007/s11123-019-00554-9](https://doi.org/10.1007/s11123-019-00554-9).
- Azzalini, A., and A. Capitanio. 2014. *The Skew-Normal and related families*. Cambridge, UK: Cambridge University Press.
- Boyes, W. J., D. L. Hoffman, and S. A. Low. 1989. An econometric analysis of the bank credit scoring problem. *Journal of Econometrics* 40 (1):3–14. doi:[10.1016/0304-4076\(89\)90026-2](https://doi.org/10.1016/0304-4076(89)90026-2).
- Carrasco, R. 2001. Binary choice with binary endogenous regressors in panel data: Estimating the effect of fertility on female labor participation. *Journal of Business & Economic Statistics* 19 (4):385–94. doi:[10.1198/07350010152596637](https://doi.org/10.1198/07350010152596637).
- Cox, D. R., and N. Wermuth. 1991. A simple approximation for bivariate and trivariate normal integrals. *International Statistical Review / Revue Internationale de Statistique* 59 (2):263–69. doi:[10.2307/1403446](https://doi.org/10.2307/1403446).
- Drezner, Z., and G. O. Wesolowsky. 1990. On the computation of the bivariate normal integral. *Journal of Statistical Computation and Simulation* 35 (1–2):101–107. doi:[10.1080/00949659008811236](https://doi.org/10.1080/00949659008811236).
- Elderton, E. M., M. Moul, E. C. Fieller, S. J. Pretorius, and A. E. R. Church. 1930. On the remaining tables for determining the volumes of a bivariate normal surface. *Biometrika* 22:13–35. Introduction by K. Pearson, 1–12.
- Evans, W. N., and R. M. Schwab. 1995. Finishing high school and starting college: Do Catholic schools make a difference? *Quarterly Journal of Economics* 110 (4):941–74. doi:[10.2307/2946645](https://doi.org/10.2307/2946645).
- Freedman, D. A., and J. S. Sekhon. 2008. Endogeneity in probit response models. Working paper.
- Genz, A. 2004. Numerical computation of rectangular bivariate and trivariate normal and t probabilities. *Statistics and Computing* 14 (3):251–60. doi:[10.1023/B:STCO.0000035304.20635.31](https://doi.org/10.1023/B:STCO.0000035304.20635.31).
- Greene, W. H. 2000. *Econometric analysis*, 4th ed. New York, NY: Prentice Hall.
- Kotz, S., N. Balakrishnan, and N. L. Johnson. 2000. Bivariate and trivariate normal distribution. Chapter 46. In *Continuous multivariate distributions, volume 1: Models and applications*. Hoboken, NJ: John Wiley & Sons.
- Lee, A. 1917. Further supplementary tables for determining high (tetrachoric) correlations from tetrachoric groupings. *Biometrika* 11 (4):284–91. doi:[10.2307/2331827](https://doi.org/10.2307/2331827).
- Lee, A. 1927. Supplementary tables for determining correlation from tetrachoric groupings (tetrachoric correlations). *Biometrika* 19 (3–4):354–404. doi:[10.1093/biomet/19.3-4.354](https://doi.org/10.1093/biomet/19.3-4.354).

- Lin, J.-T. 1995. A simple approximation for the bivariate normal integral. *Probability in the Engineering and Informational Sciences* 9 (2):317–21. doi:[10.1017/S0269964800003880](https://doi.org/10.1017/S0269964800003880).
- Owen, D. B. 1956. Tables for computing bivariate normal probabilities. *The Annals of Mathematical Statistics* 27 (4):1075–90. doi:[10.1214/aoms/1177728074](https://doi.org/10.1214/aoms/1177728074).
- Pearson, K. 1901. Mathematical contributions to the theory of evolution-VII. On the correlation of characters not quantitatively measurable. *Philosophical Transactions of the Royal Society of London, Series A* 195:1–47.
- Pearson, K. 1903. Mathematical contributions to the theory of evolution-XI. On the influence of natural selection on the variability and correlation of organs. *Philosophical Transactions of the Royal Society of London, Series A* 200:1–66.
- Pearson, K. 1931. *Tables for Statisticians and Biometricians*, Vol. 2. London, UK: Cambridge University Press.
- Schervish, M. J. 1984. Multivariate normal probabilities with error bound. *Applied Statistics* 33 (1):81–94. doi:[10.2307/2347670](https://doi.org/10.2307/2347670).
- Terza, J. V., and U. Welland. 1991. A comparison of bivariate normal algorithms. *Journal of Statistical Computation and Simulation* 39 (1–2):115–27. doi:[10.1080/00949659108811343](https://doi.org/10.1080/00949659108811343).
- Tsay, W. J., C. J. Huang, T.-T. Fu, and I.-L. Ho. 2013. A simple closed-form approximation for the cumulative distribution function of the composite error of stochastic frontier models. *Journal of Productivity Analysis* 39 (3): 259–69. doi:[10.1007/s11123-012-0283-1](https://doi.org/10.1007/s11123-012-0283-1).
- Wooldridge, J. M. 2002. *Econometric analysis of cross section and panel data*. Cambridge, MA: MIT Press.